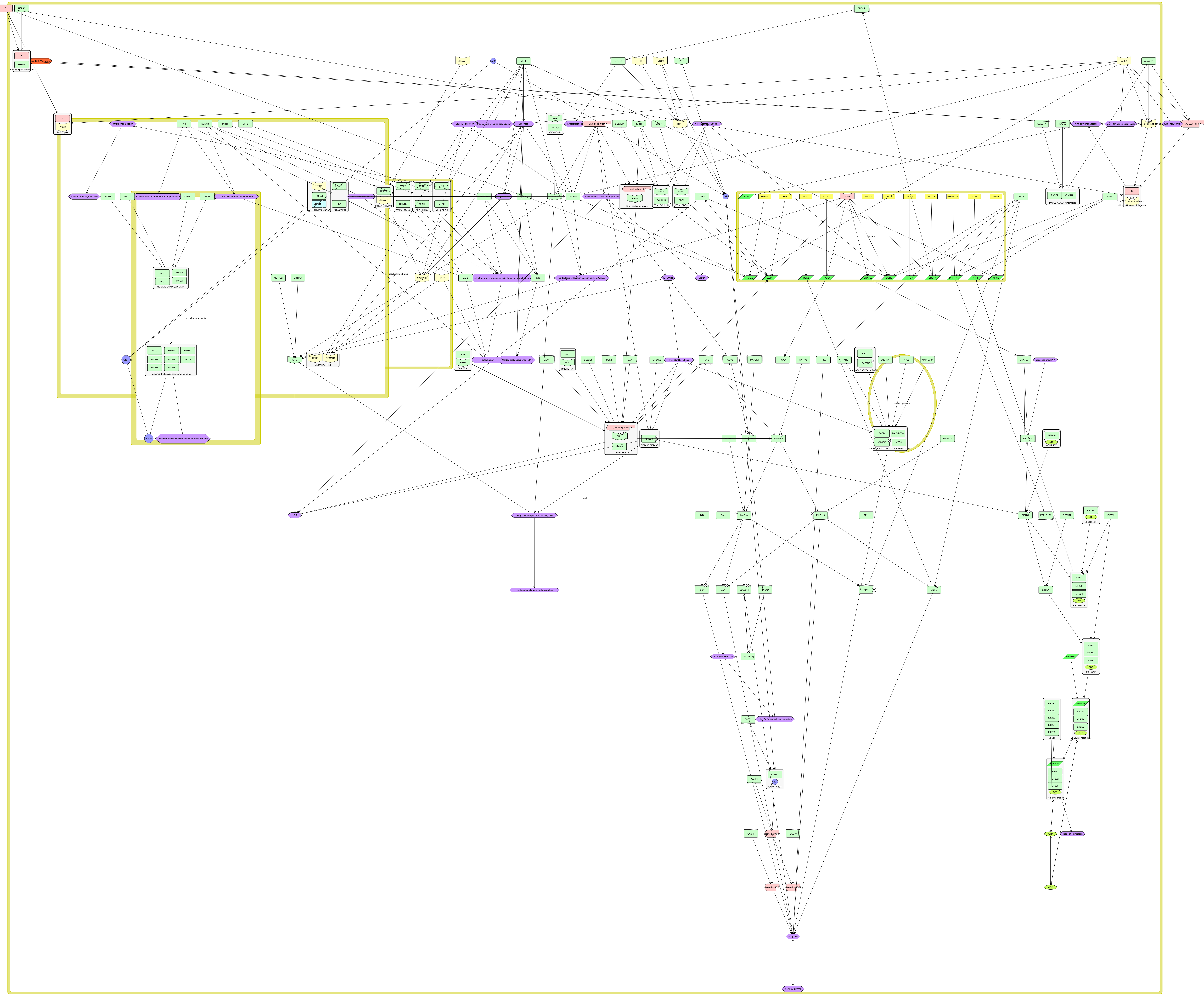
ACE2, soluble
SARS-CoV infection
S
S
S
pulmonary fibrosis
S
ACE2, membrane-bound
active
ACE2-SARS-CoV interaction
active
ACE2, membrane-bound
ADAM17
ACE2
viral RNA genome replication
S
ACE2
ACE2:Spike
HSPA5
S
HSPA5
HSPA5-Spike interaction
viral entry into host cell
PACS2
ADAM17
PACS2-ADAM17 interaction
PACS2
ADAM17
early endosome
ATF6
MFN2
HSPA5
ACE2
XBP1
BCL2
DDIT3
TRIB3
ATF4
ERO1A
HYOU1
PPP1R15A
DNAJC3
MFN2
HSPA5
ACE2
XBP1
BCL2
DDIT3
TRIB3
ATF4
ERO1A
HYOU1
PPP1R15A
DNAJC3
nucleus
&
P
DDIT3
DDIT3
XBP1
TRIB3
ATF4
ERO1A
HYOU1
PPP1R15A
DNAJC3
MAPK14
Persistant ER Stress
cleaved~CASP3
CASP3
cleaved~CASP4
CASP4
cleaved~CASP9
CASP9
CAPN1
Ca2+
CAPN1:Ca2+
CAPN1
Ca2+
Ca2+
DNAJC3
presence of dsRNA
EIF2AK2
EIF2AK2
EIF2AK4
ATP
GCN2:ATP
P
EIF2S1
EIF2S1
EIF2AK1
P
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2-P:GDP
EIF2S2
EIF2S3
GDP
EIF2S3:GDP
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2:GDP
Met-tRNA
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2:GDP:Met-tRNA
GTP
GDP
GTP
GDP
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GTP
Ternary Complex
EIF2B1
EIF2B2
EIF2B3
EIF2B4
EIF2B5
EIF2B
Translation initiation
high Ca2+ cytosolic concentration
release of ER Ca2+
XBP1
ERAD
protein ubiquitination and destruction
MFN2
endoplasmic reticulum organization
Apoptosis
ERN1
BCL2L11
ERN1:BCL2L11
ERN1
BBC3
ERN1:BBC3
BCL2L11
ERN1
BBC3
ERO1A
hyperoxidation
SIGMAR1
endoplasmic reticulum
TMBIM6
ITPR
ITPR
RYR1
ERO1A
BAX
ERN1
BAX:ERN1
accumulation of misfolded protein in ER
ER Stress
Persistant ER Stress
Unfolded protein
retrograde transport from ER to cytosol
BAX
BAK1
BCL2
unfolded protein response (UPR)
ER stress
mitochondrion-endoplasmic reticulum membrane tethering
endoplasmic reticulum calcium ion homeostasis
HSPA5
p20
PACS2
autophagy
BCAP31
VAPB
Ca2+
Ca2+ ER depletion
Unfolded protein
BCL2L1
Unfolded protein
Unfolded protein
ERN1
ERN1:Unfolded protein
ERN1
Unfolded protein
P
ERN1
TRAF2
TRAF2:ERN1
BAK1
ERN1
BAK1:ERN1
EIF2AK3
P
P
EIF2AK3
EIF2AK3:EIF2AK3
ATF6
ATF6
HSPA5
ATF6:HSPA5
HSPA5
Unfolded protein
MFN2
MFN1
MFN1:MFN2
MFN2
MFN2
MFN2:MFN2
VAPB
RMDN3
VAPB:RMDN3
mitochondria-associated endoplasmic reticulum membrane
Ca2+ cytosolic concentration
BCAP31
FIS1
FIS1:BCAP31
ITPR3
HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
ITPR3
SIGMAR1
SIGMAR1:ITPR3
ITPR3
SIGMAR1
HSPA5
SIGMAR1
SIGMAR1:HSPA5
MFN1
MFN2
RMDN3
FIS1
mitochondrion
mitochondrial fission
mitochondria fragmentation
Ca2+
Ca2+
MCU
SMDT1
SMDT1
MICU1
MICU2
MCUb
MICU1
MICU2
Mitochondrial calcium uniporter complex
mitochondrial matrix
mitochondrial outer membrane depolarization
Ca2+ mitochondrial concentration
Ca2+
mitochondrial calcium ion transmembrane transport
MCU
SMDT1
MCU
SMDT1
MCU1
MCU2
MCU:MICU1:MICU2:SMDT1
MCU2
MCU1
TRAF2
TRAF2
MAP3K5
P
MAP3K5
MAPK14
P
MAPK14
CDK5
?
CDK5
MAP3K4
P
MAP3K4
MAPK8
P
MAPK8
AP-1
P
P
AP-1
DDIT3
P
DDIT3
BAX
P
BAX
BID
P
BID
BCL2L11
P
BCL2L11
PPP2CA
cell
Cell survival
Apoptosis
TRIB3
FADD
CASP8
K63
Ub
CASP8:CASP8-ubq:FADD
SQSTM1
MAP1LC3A
ATG5
FADD
MAP1LC3A
CASP8
K63
Ub
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
TRIM13
Golgi apparatus
UPR
XBP1
ATF6
ATF6
MBTPS1
MBTPS2
ATF6


S
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HSPA5
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HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
BCAP31
FIS1
FIS1:BCAP31
Ca2+ cytosolic concentration
HSPA5
SIGMAR1
SIGMAR1:HSPA5
VAPB
RMDN3
VAPB:RMDN3
MFN2
MFN1
MFN1:MFN2
MFN2
MFN2
MFN2:MFN2
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ERN1:BCL2L11
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MFN2
nucleus
DDIT3
PACS2
ADAM17
PACS2-ADAM17 interaction
ATF4
S
active
ACE2, membrane-bound
ACE2-SARS-CoV interaction
MCU
SMDT1
MCU1
MCU2
MCU:MICU1:MICU2:SMDT1
MBTPS2
MBTPS1
endoplasmic reticulum membrane
SIGMAR1
ITPR3
VAPB
mitochondrion-endoplasmic reticulum membrane tethering
p20
endoplasmic reticulum calcium ion homeostasis
ER Stress
ERAD
HSPA5
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BCL2
HYOU1
DNAJC3
DDIT3
TRIB3
ERO1A
PPP1R15A
ATF4
MFN2
mitochondrial matrix
Ca2+
MCU
SMDT1
SMDT1
MICU1
MICU2
MCUb
MICU1
MICU2
Mitochondrial calcium uniporter complex
ATF6
ITPR3
SIGMAR1
SIGMAR1:ITPR3
BAX
ERN1
BAX:ERN1
autophagy
Unfolded protein response (UPR)
BAK1
BAK1
ERN1
BAK1:ERN1
BCL2L1
BCL2
BAX
EIF2AK3
Persistant ER Stress
TRAF2
CDK5
MAP3K4
HYOU1
MAP3K5
TRIB3
TRIM13
FADD
CASP8
Ub
CASP8:CASP8-ubq:FADD
SQSTM1
ATG5
MAP1LC3A
DNAJC3
presence of dsRNA
autophagosome
Ca2+
mitochondrial calcium ion transmembrane transport
Unfolded protein
P
ERN1
TRAF2
TRAF2:ERN1
EIF2AK3
P
P
EIF2AK3:EIF2AK3
MAPK8
MAP3K4
P
MAP3K5
P
FADD
MAP1LC3A
CASP8
Ub
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
MAPK14
EIF2AK2
EIF2AK4
ATP
GCN2:ATP
cell
UPR
retrograde transport from ER to cytosol
BID
BAX
MAPK8
P
MAPK14
P
AP-1
EIF2S1
P
PPP1R15A
EIF2AK1
EIF2S3
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EIF2S3:GDP
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BAX
P
BCL2L11
P
PPP2CA
AP-1
P
P
DDIT3
P
EIF2S1
EIF2S1
P
EIF2S2
EIF2S3
GDP
EIF2-P:GDP
release of ER Ca2+
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EIF2B4
EIF2B5
EIF2B
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2:GDP:Met-tRNA
CASP4
CAPN1
Ca2+
CAPN1:Ca2+
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
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